NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 PRELIMINARY EXAMINATION
Higher 2

NAME	
SUBJECT CLASS	2ma2
REGISTRATION NUMBER	

MATHEMATICS 9758/01

Paper 1 **1 September 2026**

3 hours

Candidates answer on the Printed Answer Booklet.

Additional Materials: Printed Answer Booklet
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Write your name, class and registration number on the work you hand in.
Write in dark blue or black pen.

Answer **all** the questions.
Write your answers on the Printed Answer Booklet. Follow the instructions on the front cover of the answer booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are **not** allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.
You must show all necessary working clearly.

You are reminded of the need for clear presentation in your answers. Up to 2 marks may be deducted for improper presentation.

The number of marks is given in brackets [ ] at the end of each question or part question.

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1 The complex numbers z and w satisfy the following equations.

$$zw^* = 5 - 5\mathrm{i}$$
$$z - 2w = 1 - 3\mathrm{i}$$

Solve the simultaneous equations, giving your answers in the form $x + \mathrm{i}y$ where x and y are real numbers. [5]

2 The function $y = \mathrm{f}(x)$ is such that $\mathrm{f}'(x) = -\dfrac{ax}{\left(x^2 - b\right)^2}$, where a and b are positive constants and $x \neq \pm\sqrt{b}$.

(a) Show that $\mathrm{f}(x) = \dfrac{a}{2\left(x^2 - b\right)} + c$, where c is an arbitrary constant. [1]

(b) In the case where the equation $\mathrm{f}(x) = 0$ has two real and distinct roots and c is positive, sketch the graph of $y = \mathrm{f}(x)$. Label clearly, in terms of a, b and/or c, the equations of any asymptotes, and the coordinates of the points where the graph crosses the axes. [4]

(c) State the range of values of c for which the graph of $y = \mathrm{f}(x)$ does not intersect the x-axis, giving your answer in terms of a and/or b. [1]

3 **(a)** By writing $\mathrm{e}^{2x} = \mathrm{e}^x\left(\mathrm{e}^x + 1\right) - \mathrm{e}^x$ or otherwise, find $\displaystyle\int \frac{\mathrm{e}^{2x}}{\left(\mathrm{e}^x + 1\right)^2}\,\mathrm{d}x$. [2]

(b) The curve C has equation $y = \dfrac{\mathrm{e}^x}{\mathrm{e}^x + 1}$, for $x \in \mathbb{R}$.

(i) The curve is translated $\dfrac{3}{4}$ units in the negative y-direction. Write down its corresponding equation. [1]

The region R is bounded by C, the y-axis, and the line $y = \dfrac{3}{4}$.

(ii) Hence, show that the exact volume of the solid formed when R is rotated through 2π radians about the line $y = \dfrac{3}{4}$ can be expressed as $\pi\left(a + b\ln 2 + c\ln 3\right)$, where a, b and c are rational constants to be determined. [4]

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4 With reference to the origin O, the points A and B have position vector $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $\mathbf{a}$ is a unit vector, $|\mathbf{b}| = 4$ and $|\mathbf{a} - \mathbf{b}| = \sqrt{13}$.

(a) By considering $(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$, find the exact value of $\angle AOB$. [3]

(b) Hence, find the exact value of $|(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - 2\mathbf{b})|$. [3]

5 Three complex numbers z_1, z_2 and z_3 are the roots of the equation

$$\mathrm{f}(z) = z^3 + az^2 + bz + c = 0,$$

where a, b and c are real constants. It is given that the product of all three roots is $-5\sqrt{5}$.

(a) By considering the product of the linear factors of $\mathrm{f}(z)$, show that $c = 5\sqrt{5}$. [1]

(b) Given further that $z_1 = 1 + 2\mathrm{i}$, find the other roots of the equation. [3]

The points representing z_1, z_2 and z_3 are denoted by Z_1, Z_2 and Z_3 on an Argand diagram such that Z_1, Z_2 and Z_3 are in a clockwise order about the origin.

(c) State the radius of the circle passing through Z_1, Z_2 and Z_3. [1]

(d) Hence state the exact values of p and q such that

$$\arg(z_1) - \arg(z_2) = 2\tan^{-1} p = 4\tan^{-1} q.$$ [2]

6 A convergent geometric progression G has first term a and common ratio r, where a and r are non-zero. It is given that the first and third terms of G are equal to the first and third terms of an arithmetic progression A respectively. The second term of A is $\frac{5}{4}$ times the second term of G.

(a) Show that $2r^2 - 5r + 2 = 0$ and hence find the common ratio of G. [4]

(b) Given that the sum of the first three terms of A is 15, find the sum to infinity of G. [2]

Let G_n denote the nth term of G.

(c) Find the exact value of $\sum_{n=1}^{\infty} \frac{G_n}{(n-1)!}$. [2]

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7 The curve C is defined parametrically by equations

$$x = 2\tan\theta, \qquad y = \cos^2\theta, \qquad \text{where } -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

(a) Without the use of a calculator, find the set of values of θ for which the curve has positive gradient. [3]

A point $P\left(2\tan p, \cos^2 p\right)$ moves on C such that p increases at a constant rate of 6 units per second.

(b) Let $g = \frac{\mathrm{d}y}{\mathrm{d}x}$. Find exactly the rate at which g is increasing at the instant when $p = \frac{\pi}{4}$. [4]

The normal to the curve at P meets the x-axis at point N.

(c) Find an expression for the area of triangle OPN in terms of p, where $p \neq 0$. What can be said about the area of triangle OPN as $p \to \frac{\pi}{2}$? Justify your answer. [4]

8 Use the substitution $x = a\sin\theta$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, to show that

$$\int \sqrt{a^2 - x^2}\ \mathrm{d}x = \frac{1}{2}x\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$$

where a is a positive constant. [4]

Hence, find

(a) the exact value of $\int_0^1 \left|\sqrt{1 - x^2} - \frac{1}{2}\right| \mathrm{d}x$, [4]

(b) $\int x\sin^{-1} x\ \mathrm{d}x$. [4]

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9 The function g is defined by

$$\mathrm{g}(x) = \begin{cases} -\sqrt{4-x^2}, & \text{for } -2 \le x \le 2, \\ \dfrac{5}{3}x - \dfrac{10}{3}, & \text{for } 2 < x < 5, \end{cases}$$

and that $\mathrm{g}(x) = \mathrm{g}(x+7)$ for all real values of x.

(a) Sketch the graph of $y = \mathrm{g}(x)$ for $-7 \le x \le 11$. [3]

(b) Find the total area of the regions between the graph of $y = \mathrm{g}(x)$ and the x-axis from $x = -7$ to $x = -2$. [2]

Suppose the domain of g is further restricted to $-5 < x \le k$, where k is a constant.

(c) State the greatest value of k such that the function g^{-1} exists. [1]

(d) Using the greatest value of k from part **(c)**, find $\mathrm{g}^{-1}(x)$ and state its domain. [4]

(e) The function f is given by $\mathrm{f}(x) = \sin x$, $0 \le x \le 2\pi$.

Solve the equation $\mathrm{g}^{-1}\mathrm{f}(x) = -\dfrac{\sqrt{13}}{2}$ exactly. [2]

10 An autocatalytic reaction is a type of a chemical reaction where an autocatalyst speeds up its own production from a substrate. In such a reaction, the total combined amount of autocatalyst and substrate remains constant throughout the reaction.

In a first order autocatalytic reaction, the rate at which the autocatalyst is produced is directly proportional to the product of the amount of the autocatalyst present and the amount of the substrate present with a positive constant of proportionality k.

Let A denote the amount of autocatalyst present at time t seconds after the start of the reaction and the constant C denote the total combined amount of autocatalyst and substrate present.

Initially, the amount of autocatalyst present is $\dfrac{C}{5}$ and the rate at which the autocatalyst is produced is $\dfrac{2C^2}{25}$.

(a) Show that $\dfrac{\mathrm{d}A}{\mathrm{d}t} = \dfrac{A}{2}(C - A)$. [3]

(b) Hence, solve the differential equation to find A in terms of t and C. [5]

(c) Sketch the graph of A against t. [2]

(d) Given that the maximum rate at which the autocatalyst is produced occurs 10 seconds after the start of the reaction, find the value of C. [2]

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11 Roger and Rafael are playing tennis. The origin O is taken to be a point on the ground below Roger. The x-axis points down the court, the y-axis points across the court to Roger's left, and the z-axis points vertically upwards, as shown in Fig 1. The ball is modelled as a point.

In this question, the ball moves sufficiently fast such that it may be assumed that the ball only travels in straight lines.

At the instant of contact, Roger's racket face is modelled by a plane π_1 which contains the point $A(0.05, -0.30, 2.7)$. The ball leaves the racket in a direction $6\mathbf{i}+\mathbf{j}-\mathbf{k}$ perpendicular to the racket face, as shown in Fig. 2. Shortly after contact, the ball is at point $P(3, 0.5, 2.2)$.

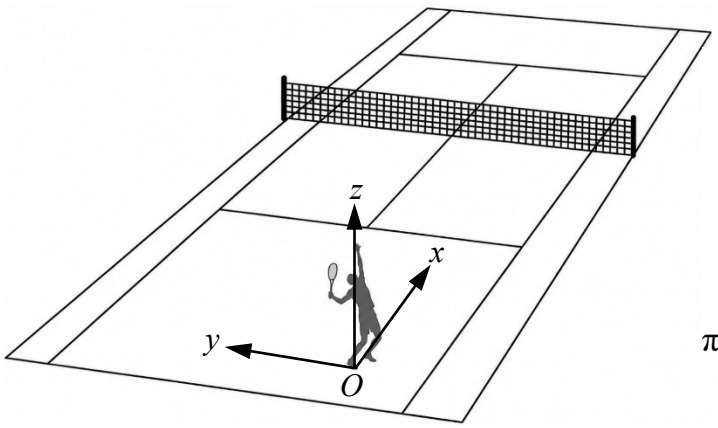


Fig. 1

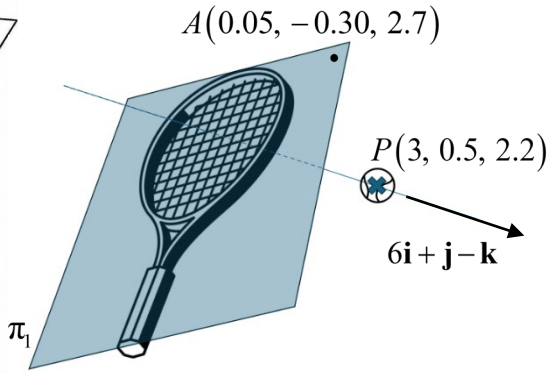


Fig. 2

(a) Find an equation of π_1 in scalar product form. [2]

(b) Find the coordinates of the point of contact of the ball with Roger's racket face. [3]

The ball hits the ground and bounces towards Rafael's racket along the line m which contains the point $B(16.2, 2.7, 0)$ and is parallel to $18\mathbf{i}+3\mathbf{j}+2\mathbf{k}$. The ball then hits Rafael's racket face, modelled by the plane π_2, and rebounds along the line m' which is the reflection of the line m in π_2. The point where the ball hits Rafael's racket face is denoted by Q. It may be assumed that the lines m, m' and the normal to π_2 at Q all lie in the same plane.

Suppose π_2 has equation $36x+5y-z=D$.

(c) Given that the z-coordinate of Q is 0.9, find the value of D. [2]

(d) Find the projection vector of $\overrightarrow{BQ}$ onto the normal of π_2. [3]

(e) Hence, find a vector equation of the line m'. [2]

Suppose instead that π_2 has equation $ax+by+cz=d$.

(f) Given that the ball hits Rafael's racket perpendicular to the racket face, express each of a and b in terms of c. [2]

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